In constant currency, total margin and total margin as a percentage of revenues for our hardware segment increased in fiscal 2019 due to lower expenses .

Services Business

We offer services to customers and partners to help to maximize the performance of their investments in Oracle applications and infrastructure technologies. Services revenues are generally recognized as the services are performed. The cost of providing our services consists primarily of personnel related expenses, technology infrastructure expenditures, facilities expenses and external contractor expenses.

	Year Ended May 31,			
		Percent Change		
(Dollars in millions)	2019	Actual	Constant	2018
Services Revenues:				
Americas	$ 1,576	-5%	-3%	$ 1,654
EMEA	1,021	-2%	2%	1,046
Asia Pacific	643	-7%	-4%	695
Total revenues	3,240	-5%	-2%	3,395
Total Expenses (1)	2,703	-1%	2%	2,729
Total Margin	$ 537	-19%	-18%	$ 666
Total Margin %	17%			20%
% Revenues by Geography:				
Americas	49%			49%
EMEA	31%			31%
Asia Pacific	20%			20%

(1) Excludes stock-based compensation and certain expense allocations. Also excludes amortization of intangible assets and certain other GAAP-based expenses, which were not allocated to our operating segment results for purposes of reporting to and review by our CODMs, as further described under "Presentation of Operating Segments and Other Financial Information" above.

Excluding the effects of currency rate fluctuations, our total services revenues decreased in fiscal 2019 relative to fiscal 2018 primarily due to revenue declines in our education services and, to a lesser extent, our consulting services. During fiscal 2019, constant currency increases in our EMEA-based services revenues were offset by constant currency services revenue decreases in the Americas and the Asia Pacific regions.

In constant currency, total services expenses increased in fiscal 2019 compared to fiscal 2018 primarily due to an increase in employee related expenses and external contractor expenses associated with investments in our consulting services that support our cloud offerings. In constant currency, total margin and total margin as a percentage of total services revenues decreased during fiscal 2019 relative to fiscal 2018 due to decreased revenues and increased expenses for this business.

Research and Development Expenses: Research and development expenses consist primarily of personnel related expenditures. We intend to continue to invest significantly in our research and development efforts because, in our judgment, they are essential to maintaining our competitive position.

	Year Ended May 31,			
		Percent Change		
(Dollars in millions)	2019	Actual	Constant	2018
Research and development (1)	$ 5,063	-2%	0%	$ 5,163
Stock-based compensation	963	5%	5%	921
Total expenses	$ 6,026	-1%	0%	$ 6,084
% of Total Revenues	15%			15%

(1) Excluding stock-based compensation

On a constant currency basis, total research and development expenses were flat in fiscal 2019, as lower employee related expenses including lower variable compensation were offset by an increase in stock-based compensation expenses .

General and Administrative Expenses: General and administrative expenses primarily consist of personnel related expenditures for IT, finance, legal and human resources support functions; and professional services fees.

	Year Ended May 31,			
		Percent Change		
(Dollars in millions)	2019	Actual	Constant	2018
General and administrative (1)	$ 1,093	-1%	2%	$ 1,102
Stock-based compensation	172	-5%	-5%	180
Total expenses	$ 1,265	-1%	1%	$ 1,282
% of Total Revenues	3%			3%

(1) Excluding stock-based compensation

Excluding the effects of currency rate fluctuations, total general and administrative expenses increased in fiscal 2019 compared to fiscal 2018 primarily due to increased professional services fees.

Amortization of Intangible Assets: Substantially all of our intangible assets were acquired through our business combinations. We amortize our intangible assets over, and monitor the appropriateness of, the estimated useful lives of these assets. We also periodically review these intangible assets for potential impairment based upon relevant facts and circumstances. Note 6 of Notes to Consolidated Financial Statements included elsewhere in this Annual Report has additional information regarding our intangible assets and related amortization.

	Year Ended May 31,			
		Percent Change		
(Dollars in millions)	2019	Actual	Constant	2018
Developed technology	$ 857	13%	14%	$ 758
Cloud services and license support agreements and related relationships	712	-3%	-3%	731
Other	120	-9%	-9%	131
Total amortization of intangible assets	$ 1,689	4%	4%	$ 1,620

Amortization of intangible assets increased in fiscal 2019 compared to fiscal 2018 primarily due to additional amortization from intangible assets, which primarily included developed technology that we acquired in connection with our recent acquisitions, partially offset by a reduction in expenses associated with certain of our intangible assets that became fully amortized.

Acquisition Related and Other Expenses: Acquisition related and other expenses consist of personnel related costs and stock-based compensation for transitional and certain other employees, integration related professional services, and certain business combination adjustments including certain adjustments after the measurement period has ended and certain other operating items, net. Stock-based compensation expenses included in acquisition related and other expenses resulted from unvested restricted stock-based awards and stock options assumed from acquisitions whereby vesting was accelerated generally upon termination of the employees pursuant to the original terms of those restricted stock-based awards and stock options.

	Year Ended May 31,			
		Percent Change		
(Dollars in millions)	2019	Actual	Constant	2018
Transitional and other employee related costs	$ 49	3%	4%	$ 48
Stock-based compensation	—	-100%	-100%	1
Professional fees and other, net	16	373%	426%	3
Business combination adjustments, net	(21)	*	*	—
Total acquisition related and other expenses	$ 44	-15%	-13%	$ 52

* Not meaningful